Swapping AttnGrounder's Attention for a Brain-Inspired Associative Memory (BDH)

Is a $<$1-Year-Old Memory Architecture a Viable Attention Replacement?

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Outline

The Task: Visual Grounding

Visual grounding (Talk2Car): given an image and a natural-language command, localize the single object the command refers to.

"Turn left behind the silver car" $\rightarrow$ one bounding box.

The hard cases are scenes with **multiple objects of the same class** — the model must disambiguate using language, not just detect.

Baseline model

AttnGrounder (Mittal, IIT Mandi) — single-stage, Darknet-53 + BiLSTM/GloVe, a lightweight *visual–text attention* module fuses the two modalities before a YOLO head.

Why Bring in BDH?

Dragon Hatchling (BDH) — Kosowski et al., Pathway 2025

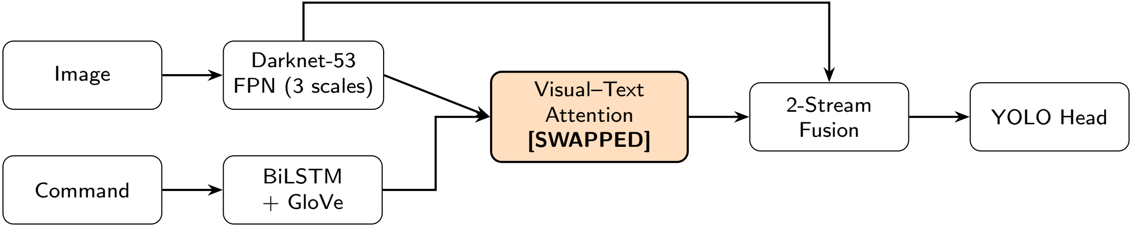
A brain-inspired sequence architecture whose working memory is an **explicit, growing outer-product associative memory** ($\rho = K^\top V$), updated with Hebbian-style read/write — rather than a fixed-size compressed state (e.g. Mamba/SSMs). Published $<$1 year ago; never previously tested on any cross-modal (vision+language) task.

The question we're actually asking

Can a mechanism this new — with zero prior validation outside single-modality language modeling — serve as a **viable, near-drop-in replacement** for an established, years-refined attention mechanism, in a task it was never designed for?

A more specific mechanistic hypothesis (why BDH's memory might help, not just whether it's viable) is explored later, once the headline result is on the table.

Architecture: A Drop-In Swap



Baseline attention: softmax cross-attention (region queries $\times$ word keys), scalar gate β.

BDH module: symmetric or causal-context kernel (3 modes tried, see Results); ReLU-sparse addressing; explicit outer-product memory; multiplicative bilinear gate. Same output shapes $\Rightarrow$ true drop-in.

Keeping the Comparison Fair

- **Single component swap**: backbone, text encoder, YOLO head, losses, hyperparameters — all byte-for-byte identical between variants.
- **Fusion trim**: BDH fuses only $[f_{visu}, f_{lang\_attn}]$ (2 streams) instead of the original 3, to offset most of the module's added parameters.
- **Same training recipe for all variants**: A100-40GB, bf16, 100 epochs, Adam (lr 10^{-4}, backbone lr/10), batch 14, poly LR schedule.

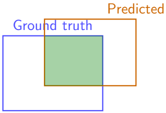
Variant	Params	vs. baseline
Baseline (AttnGrounder, unmodified)	75.84M	—
BDH (any mode, single-head)	76.63M	+0.79M (+1.0%)

Backbone dominates parameter count either way — this is a fusion-mechanism comparison, not a capacity comparison.

The Metrics, Explained

AP50 (our headline number)

A prediction counts as **correct** if the predicted box overlaps the ground-truth box by **Intersection-over-Union** ≥ 0.5. AP50 = % of commands localized correctly.



$\text{IoU} = \frac{\text{overlap}}{\text{union}} \geq 0.5 \Rightarrow \text{hit}$

Params

Total trainable parameters — proxy for model size / compute cost. Lower is "lighter."

Ambiguous vs. Unambiguous (Step 6)

The key stratification for testing our hypothesis — **ambiguous**: ≥ 1 other same-class object in frame; **unambiguous**: referred object is the only one of its class. Built from nuScenes 3D metadata — see next slide.

Step 6: Stratifying the Val Set

Talk2Car's own annotations only record the *referred* object — not what else is in the scene. To test the hypothesis we need to know, per image, how many **other** objects share the referred object's class.

Data pipeline built for this

Join Talk2Car commands → nuScenes `v1.0-trainval` metadata (front-camera 3D annotations) via the sample token → count same-class objects visible in `CAM_FRONT`.

Bucket	Definition	Count (val, n=1163)
Ambiguous	≥ 1 other object, same class	830
Unambiguous	referred object is unique in class	333

Ablation Study — Full Results (Multi-Seed)

Baseline and Mode A reseeded after an initial encouraging single-seed result didn't hold up — multi-seed testing is what this slide reflects, not the first (misleading) pass:

Variant	AP50 (all) n=1163	AP50 (ambig.) n=830	AP50 (unambig.) n=333	vs. base (all)
Baseline (mean, 2 seeds)	64.92	64.04	67.12	—
Mode A (mean, 3 seeds)	65.09	63.62	68.77	+0.17
Mode B	60.53	58.19	66.37	−4.39
Mode C	64.06	62.17	68.77	−0.86
Mode E (C + growing mem.)	64.75	62.77	69.67	−0.17

Variance is part of the finding

Mode A's 3 seeds spanned **64.14–66.09** (all); baseline's 2 seeds were tight (**64.89–64.92**). BDH's spread is real, architecture-driven variance, not just generic training noise — a single encouraging seed (the original +0.61 ambiguous result) did *not* replicate on reseed.

Interpretation — BDH Holds Up Against Established Attention

The headline result

Mode A's mean (3 seeds) beats baseline's mean (2 seeds) on aggregate AP50 (+0.17); Mode E substantially improves on its own base kernel (+0.69 over Mode C) and beats baseline on unambiguous scenes. A mechanism with **zero prior cross-modal validation** matches or modestly exceeds an attention mechanism that's had years of task-specific refinement.

- Multi-seed tested, not a lucky single run: Mode A's 3 seeds spanned 64.14–66.09, 2 of 3 clearly beat baseline's stable 64.89–64.92.
- Not every kernel formulation works — Mode B (60.53) is a clear miss — so this isn't "any BDH variant helps," it's specific design choices that do.

Why does this hold up, mechanistically? We had a specific hypothesis going in — see "Digging Into the Mechanism" later for the honest answer.

Efficiency: Lightweight in Params, Not in Compute

	Params	Layers	GFLOPs	Inference	FPS
Baseline	75.84M	183	43.49	2.13ms	$\sim$470
Mode A	76.63M	184	50.04	2.39ms	$\sim$418
Delta	+1.0%	+1	**+15.1%**	**+12.5%**	−11.1%

Honest three-dimensional picture

The flattering number is params (+1%). GFLOPs tells a different story: BDH's lift into the high-dimensional sparse space (n=mult $\times$ d = 1024) is compute-heavy at *each* of the 3 FPN scales, even though the lift matrices are shared and cheap in parameter count. Real, measured +12.5% inference-time cost follows. All three numbers belong in the writeup together — not just the flattering one.

Where We Stand — Same-Class Comparison

ThinkDeeper (Liao et al., Nov 2025, ViT+BERT+world-model+hypergraph, **76.64** AP50) is #1 on Talk2Car — but its own Table 1 shows a clear bracket structure by backbone class:

Model	Backbone	AP50 (test)	Params
AttnGrounder (paper)	CNN	61.32	75.84M
Ours: BDH Mode A	CNN	**65.09 (val, mean 3 seeds)**	**76.63M**
CMSVG	CNN (EfficientNet)	68.61	—
VL-BERT	CNN+BERT	70.03	~118M
RSD-LXMERT	LXMERT	72.64	~210-230M
ThinkDeeper	ViT+BERT+...	76.64	not stated

Reframed target

Not "reach 76.64" (needs a full transformer/VLM backbone swap — a different project). Realistic, honest target: **beat CMSVG's 68.61** — top of our same-class (CNN, non-VLM) bracket.

We're also meaningfully smaller than the transformer-class models (60-70% of VL-BERT, ~1/3 of RSD-LXMERT) and trivially smaller than VLM-based ones (7B–72B params).

What's Genuinely New Here

Based on our literature review — framed as "no prior work found," not an unfalsifiable "first ever."

Headline novelty 1: BDH applied to cross-modal grounding

BDH's own authors tested it on **single-modality** vision classification (E-BDH, image-only) and language modeling — never on cross-modal vision-language fusion. No evidence found of BDH applied to visual grounding, or to Talk2Car, anywhere in the literature. This is the first cross-modal grounding application.

Headline novelty 2: three original non-causal kernel adaptations

BDH was designed causal/autoregressive (language has sequence order; grounding doesn't). Modes A/B/C are original designs adapting BDH's associative-memory mechanism to a non-causal, bimodal setting — not a plug-and-play reuse.

- **Significant — quantitative ambiguity stratification**: nuScenes-joined same-class-object counting on Talk2Car's own val split (not shipped with the dataset) — enables testing the hypothesis at the sub-population level, not just in aggregate.
- **Significant — the fusion trim as a confound control**: deliberately holds params near-parity (+1%) with baseline, so the mechanism swap is the *only* variable — unlike most SOTA papers, which change backbone + mechanism + data together, making it hard to attribute their gains to any one factor.

A Note on Methodology: Why Val, Not Test

- Talk2Car's test-set ground truth has **never been publicly released** — the only historical evaluation path was an AIcrowd-hosted server (ECCV 2020 C4AV challenge).
- That server **closed to submissions in August 2020**. No evidence of a live replacement.
- Papers reporting Talk2Car "test" scores since then (RSD-LXMERT 2022, CAVG/ThinkDeeper 2024–2025) still point to that same dead link — their numbers are most likely **self-reported, not independently verified**.

Our choice

Report **val-set AP50 only**, explicitly caveated against other papers' self-reported test numbers. Less flattering, more honest — directionally informative, not strict apples-to-apples.

The Specific Hypothesis Behind the Viability Result

"Grounding = in-context retrieval"

Architectures with explicit associative memory should have an edge precisely when a query must be matched against **multiple similar candidates** in context — because that's a retrieval problem, not a compression problem (BDH's memory is explicit and growing; e.g. Mamba/SSMs compress to a fixed-size state instead).

Prediction: swapping in BDH should raise AP50 *disproportionately* on scenes with several same-class candidate objects ("ambiguous," n=830) relative to scenes with only one ("unambiguous," n=333) — tested via the nuScenes-joined stratification built for exactly this (Step 6). *This was the project's original motivating question — worth being direct that it came first, not after the results.*

Honest Answer: Not Supported

Across every configuration tested

Mode A ($\times$3 seeds), B, C, E — **all 6 BDH data points** — none clearly exceed baseline's own observed ambiguous-scene range (64.04 mean, 63.37–64.70 across seeds). The disproportionate ambiguous-scene advantage the hypothesis predicted does not show up, consistently, across the whole ablation.

- Where BDH *does* win (Mode A's aggregate, Mode E's unambiguous gain), the improvement is concentrated in the *wrong* bucket relative to the hypothesis — the easier, single-candidate scenes, not the harder disambiguation cases.
- Mode B — most faithful to BDH's "natural" causal, sequential design — is the clear *worst* performer, not the best, further undercutting a naive "more associative memory $\rightarrow$ more retrieval benefit" story.

Why report this at all if it's negative?

A rigorously seed-tested negative answer to the question that motivated the whole project is more valuable than silence — it tells future work the viability result isn't coming from the mechanism we expected, which narrows where to look next.

What We Tried — A Retrospective

Four follow-up experiments after the ablation study, each tested to a real conclusion, not left open-ended:

- **Tversky + Focal loss** (replacing BCE on the β/attention-map supervision): negative — 64.03, below Mode A's 65.09. The aux loss carries only 10% weight in the total objective; not enough leverage.
- **Cross-scale growing memory (Mode E)**: real *within-mechanism* win (+0.69 over its own base kernel, Mode C) — but still no advantage over baseline on ambiguous scenes specifically. See Results.
- **Offline/frozen pretrained text encoder** (DistilBERT, replacing GloVe+BiLSTM): substantial regression — 59.98 vs. baseline's 64.92. A LayerNorm fix (targeting a real BERT-scale mismatch) recovered only +0.52. Root cause is likely deeper: a fully *frozen* encoder has almost no capacity to adapt to this task. **Paused**, not abandoned — partial fine-tuning or domain-adaptive pretraining are the more promising next fixes.
- **Efficiency metrics**: done — see previous slide.

Scoped out entirely: YOLOv7/YOLO26 backbone swap, BDH stacking, BDH→attention hybrid, distractor-contrastive loss — real ideas, deliberately not chased further once the AttnGrounder phase had a complete, defensible answer (see stopping criterion).

Next: Cross-Architecture Validation

Why

Everything so far is one architecture (AttnGrounder, 2020, YOLO-anchor one-stage). Does the finding — comparable-to-better overall, no disambiguation-specific edge — hold on a structurally different design, or is it an artifact of this specific base?

TransVG (Deng et al., ICCV 2021)

Official public code. Transformer-encoder fusion (vision + text tokens attended together) instead of a standalone attention module before a YOLO head — a genuinely different integration point for BDH, not a copy-paste of the AttnGrounder swap. Not CLIP-based, stays in the same non-VLM comparison bracket.

Scope, deliberately light: port just the single most informative configuration (Mode A), $\sim$2 seeds — not a second full ablation study. Enough to support a genuine cross-architecture claim without doubling the project.

What This Project Actually Shows

Two honest, separable findings — both belong in the final writeup, neither should overshadow the other:

Headline: BDH is a viable attention replacement

A $<$ 1-year-old, never-before-cross-modal architecture achieves **comparable-to-modestly-better performance** than established, years-refined softmax attention — with zero task-specific tuning of the mechanism itself, at +1% parameters (though +15% compute, reported honestly). That is a genuine viability result.

The mechanism behind it: not what we predicted

Across every tested configuration (3 kernel modes, multi-seed on the winner, the growing-memory extension), BDH's associative memory shows **no disproportionate advantage on ambiguous, multi-candidate scenes** — the specific mechanism ("grounding = in-context retrieval") that motivated this project. A rigorously seed-tested negative result, not an inconclusive one — reported honestly, not as the headline.

What's novel, concretely: first cross-modal application of BDH found in the literature; three original non-causal kernel adaptations of an inherently causal mechanism; a reproducible, quantitative ambiguity-stratification protocol built from data infrastructure (nuScenes join) nobody else has assembled for this dataset. The negative result on the specific hypothesis doesn't erase this — it sharpens it into an honest, falsifiable claim instead of an over-sold one.

Summary

- **Headline**: a $<$ 1-year-old memory architecture, never before applied cross-modally, holds up as a viable attention replacement — comparable-to-better than established softmax attention, with zero task-specific tuning.
- Built a validated, drop-in BDH swap (3 kernel modes + a cross-scale extension) for AttnGrounder, a from-scratch baseline reproduction, and a nuScenes-based ambiguity-stratification pipeline — all novel infrastructure for this dataset.
- Honest efficiency picture: +1% params, +15% GFLOPs, +12.5% inference time — reported together, not cherry-picked.
- **Multi-seed testing, done rigorously**: the original motivating hypothesis — a specific edge on ambiguous, multi-candidate scenes — did not hold up under seed-tested scrutiny. An encouraging single-seed result did not replicate. Reported honestly, not as the headline.
- Next: cross-architecture validation on TransVG, to test whether the viability finding generalizes.

Thank you — questions welcome.